

ANUBHAVAI — VoicePath Solution Deck

"Your Experience Has Skills — Empowering Vernacular Informal Workers"

Problem Statement 04 — VoicePath Compliance Matrix

PS Requirement (04 VoicePath)	ANUBHAVAI Implementation & AI Model	Status
1. Unscripted Spoken Account Natural speech in Indian languages.	Web Speech API Live Voice Recording Real-time capture in English (en-IN), Hindi (hi-IN), or Tamil (ta-IN).	100% PASS
2. Indirect Phrasing Skill Extraction 'Fixing bikes' -> Mechanical Repair.	GPT-4o-mini + Vernacular NLP Heuristic Engine Parses informal stories into skills, confidence levels & evidence quotes.	100% PASS
3. Transcript & Skill Highlighting Live transcript display & skill badges.	Interactive Transcript & Skill Graph Displays full spoken transcript with highlighted key phrases & skill tags.	100% PASS
4. Real District-Level Scheme Dataset PM-AJAY & skilling grants matching.	PM-AJAY & PM FME District Engine Vector match scoring against PM-AJAY grants & MUDRA loan datasets.	100% PASS
5. Read Back ALOUD + Justification TTS audio narration with spoken reason.	Multilingual SpeechSynthesis TTS Narrows top matches & speaks justification grounded in user's words.	100% PASS
6. Stretch Goal (Code-Switching) Mixed Hinglish / Tanglish speech.	Code-Switched Token Parsing Extracts skills from mixed Hindi-English & Tamil-English narratives.	100% PASS

AI / ML Architecture & Model Breakdown

AI / ML Component	Model / Technique Used	Role & Responsibilities
Primary Language Model (LLM)	OpenAI GPT-4o-mini / GPT-4o (Structured JSON Mode, Temp 0.3)	Processes unscripted spoken accounts, performs indirect skill extraction, and generates scenario-based verification questions grounded in candidate experience.
Vernacular NLP Extraction Engine	Multilingual Rule-Based NLP Heuristics (20+ Occupation Regex Parsers)	Deterministic offline fallback parsing transcripts across Tailoring, Mechanical Repair, Food Production, Carpentry, Masonry, Agriculture, Electrical & Plumbing.
Code-Switching NLP Tokenizer	Vernacular Code-Switch Token Parser (Hinglish & Tanglish Rules)	Handles mixed language narratives (e.g. Hindi+English or Tamil+English) without losing skill context or evidence quotes.
Speech Recognition (STT)	Web Speech API SpeechRecognition (Locale: en-IN, hi-IN, ta-IN)	Real-time live speech-to-text recording directly inside the browser with continuous interim result streaming.
Speech Synthesis (TTS)	Web Speech API SpeechSynthesis (Native Voice Selection & Rate Control)	Reads scheme justifications, skill passport details, and recruiter voice bio audio previews aloud in native Indian languages.
Opportunity Match Engine	Cosine Similarity & Vector Match Scoring	Calculates match percentages between user skill vectors and district-level PM-AJAY, PM-FME, and MUDRA scheme requirements.

Complete Production Technology Stack

Layer	Technologies Used	Key Features & Capabilities
Frontend UI Layer	React 19, TypeScript, Vite, Vanilla CSS + Tailwind v4, Lucide Icons, ReactFlow, i18next (en/hi/ta)	Glassmorphism dark/light design, dynamic skill node graphs, localized multi-language state, PWA offline caching.
Voice & Recruiter Audio	Web Speech API, VoicePitchPlayer, Equalizer Waveform Visualizer	Recruiter Audio Pitch Player with animated equalizer bars, playback speed controls (1x to 1.5x), and native TTS voice bio.
Recruiter & Reach Portal	MSME Recruiter Portal (/recruiter), WhatsAppAlertSimulator	Direct talent search by district & skill, call/WhatsApp triggers, instant simulated WhatsApp job & PM-AJAY scheme notifications.
Backend API Layer	FastAPI (Python 3.11+), Pydantic v2, Python-Jose JWT Auth, RESTful Architecture	High-performance asynchronous endpoints for user management, skill extraction, verification quizzes, and scheme matching.
Database & Persistence	SQLAlchemy 2.0 ORM, Local MySQL (Prod) & SQLite Auto-Fallback (anubhavai.db)	Supports direct connection to local MySQL (via .env) with automatic zero-config SQLite fallback for instant hackathon demos.